Saturn

The ringed planet - sixth planet from the Sun

Overview

Saturn is the sixth planet from the Sun and the second-largest planet in the Solar System, after Jupiter. Like Jupiter, it is a gas giant composed mainly of hydrogen and helium. It is named after the Roman god of agriculture and wealth.

Saturn is best known for its spectacular ring system, the largest and brightest in the Solar System. It is also the least dense planet: its average density is lower than that of water.

Key facts

- **Average distance from the Sun:** 9.5 AU (about 1.43 billion km)
- **Diameter:** 116,460 km (about 9 times Earth)
- **Length of day:** 10 hours 33 minutes
- **Length of year:** 29.4 Earth years
- **Surface gravity:** 10.44 m/s squared (about 1.07 times Earth, at cloud tops)
- **Average temperature:** about -140 C (at cloud tops)
- **Number of moons:** the most of any planet (well over 200 known)
- **Axial tilt:** 26.7 degrees

Rings

Saturn's rings are made mostly of chunks of water ice, ranging from tiny grains to pieces as large as a house. The main rings extend up to about 282,000 km from the planet, yet in most places they are only about 10 metres to 1 kilometre thick.

The rings are named with letters (A, B, C, D, E, F and G) in the order they were discovered. The largest gap between them is the **Cassini Division**, about 4,800 km wide, between the A and B rings. Galileo first saw the rings in 1610 but could not explain them; in 1655 Christiaan Huygens correctly described them as a ring around the planet.

Atmosphere

Saturn's atmosphere is mostly hydrogen and helium. Winds near the equator can reach about 1,800 km/h. At Saturn's north pole there is a unique, long-lasting cloud pattern shaped like a **hexagon**, each side of which is wider than Earth.

Moons

Titan is Saturn's largest moon and the second-largest moon in the Solar System. It was discovered by Christiaan Huygens in 1655. Titan is the only moon with a thick atmosphere (mainly nitrogen) and the only place besides Earth known to have stable liquid lakes and seas on its surface - but they are made of liquid methane and ethane, not water.

Enceladus is a small icy moon that sprays jets of water vapour and ice from its south pole, fed by an ocean beneath its surface.

Interior

Saturn is thought to have a dense core of rock and ice surrounded by a deep layer of metallic hydrogen and an outer layer of molecular hydrogen and helium. Like Jupiter, Saturn gives off more energy than it receives from the Sun. Part of this heat may come from droplets of helium slowly 'raining' down through the hydrogen in its interior.

Saturn's magnetic field is weaker than Jupiter's but still about 580 times stronger than Earth's. Unusually, it is almost perfectly aligned with the planet's rotation axis.

Storms

Roughly once every Saturnian year (about 30 Earth years), a giant storm called a **Great White Spot** erupts in Saturn's northern hemisphere. The most recent one, in 2010-2011, was tracked in detail by the Cassini spacecraft and grew until it wrapped around the entire planet.

Age and changes of the rings

Measurements from Cassini suggest that the rings may be relatively young - perhaps only a few hundred million years old or less - although this conclusion is still debated. Material from the rings is slowly falling into Saturn, so the rings may not last forever.

Because Saturn is tilted, we see the rings from different angles during its 29-year orbit. About every 13 to 16 years the rings line up edge-on with Earth and almost disappear from view in small telescopes; this last happened in March 2025.

Other notable moons

Mimas has a huge impact crater called Herschel, which makes it look like the 'Death Star' from Star Wars. **Iapetus** has one very dark hemisphere and one very bright one. **Hyperion** has a strange sponge-like surface. Small 'shepherd moons' such as **Prometheus** and **Pandora** orbit on either side of the narrow F ring and help keep it in shape.

Exploration history

- **Pioneer 11** (NASA, 1979) - first spacecraft to fly past Saturn.
- **Voyager 1** (1980) and **Voyager 2** (1981) (NASA) - detailed images of the rings and moons.
- **Cassini-Huygens** (NASA/ESA/ASI, launched 1997, arrived 2004) - orbited Saturn for 13 years. Its Huygens probe landed on Titan on 14 January 2005, the most distant landing ever made. The mission ended on 15 September 2017 when Cassini was deliberately sent into Saturn's atmosphere ('Grand Finale').
- **Dragonfly** (NASA, planned launch 2028) - a rotorcraft that will fly between locations on Titan.

Did you know?

- Saturn would float in a bathtub big enough to hold it, because it is less dense than water.
- Saturn's rings are extremely wide but in most places thinner than a tall building.
- Titan has rain, rivers and lakes - made of methane instead of water.